Final report on the emotion classifier project

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1. Introduction

How do humans communicate?

Most people would immediately think of spoken or written language. Yet communication goes beyond words. An essential component of human interaction is *emotion*. Emotions form a key part of nonverbal communication and have been widely studied within the field of psychology.

1.1. What is facial emotion recognition

Facial emotion recognition (FER) refers to the ability to identify human emotions computationally based on images. Other types of emotion recognition tasks approach the same problem by analyzing speech or writing to reconstruct a person's emotional state. As past research has shown, understanding human emotions is no trivial task for an algorithm. The best results in the field of FER have been achieved using complex model architectures and significant amounts of training data. However, the results are quite promising. Recent publications have exceeded accuracies of 90% [Citation] in their studies.

1.2. Why is FER important

Given that humans themselves are capable of emotion recognition, it may not be immediately clear why research in this field is so important. The answer to this question is that characterizing a person's feelings is a key part of many important sectors such as:

- Healthcare
- Public safety
- Crime detection
- Advertising

[Citation]

By automating the emotion recognition process needed in these areas, the risk of human error can be significantly reduced. Furthermore individuals differ substantially in their ability to correctly classify sentiments. Implementing automated emotion recognition solutions therefore provides a consistent baseline for real-world applications.

1.3. Our work

The goal of this project was to research and understand this technology, followed by the construction, training and evaluation of our own AI model specialized in assessing six basic human emotions: happiness, anger, sadness, fear, disgust and surprise. The research done prior to the practical implementation revealed that this was already a highly researched topic with promising results not limited to one architecture or family of architectures but imporoved accuracies emerging from neural networks leveraging fundamentally different concepts.

To better understand these methods and use them to achieve the best possible results we independently implemented several promising designs, slightly altering the original parameters to compensate for the required 64×64 aspect ratio of the images. The groups final model was selected by evaluating the best performing models on a held out test set.

2. Related Work

2.1. Efficient Architectures and Low-Resolution Vision

The evolution of computer vision architectures has largely been driven by the quest for efficiency and scalability. Tan and Le introduced **EfficientNetV2** [6], highlighting the importance of training speed and parameter efficiency through the use of Fused-MBConv layers. While highly effective on ImageNet, standard CNNs often struggle with information loss when applied to small inputs. Addressing this, Hassani et al. proposed **Compact Convolutional Transformers (CCT)** [3], demonstrating that transformers can escape the "big data paradigm" by using convolutional tokenizers to handle small datasets effectively. Similarly, modern pure ConvNets like **ConvNeXt V2** [8] utilize fully convolutional masked autoencoders to improve representation learning. More recently, **Vision Mamba (Vim)** [9] challenged the transformer dominance by utilizing bidirectional State Space Models (SSMs) for linear complexity scaling, offering a computationally efficient alternative for visual representation. Our work evaluates these state-of-the-art architectures, adapting their downsampling strategies to the constraints of 64×64 facial expression recognition (FER).

2.2. Transfer Learning in Facial Analysis

While generic object recognition models provide a strong baseline, FER benefits significantly from domain-specific pre-training. Transfer learning from massive face recognition datasets (e.g., CelebA, MS-Celeb-1M) is a standard practice to overcome the data scarcity in emotion recognition. Research indicates that models pre-trained on millions of faces learn robust, identity-invariant features that are highly transferable. Wang et al. demonstrated this in their **QCS (Quadruplet Cross Similarity)** network [7], utilizing an InsightFace ResNet-50 (IR50) backbone pre-trained on MS-Celeb-1M to extract fine-grained facial features. Inspired by their finding that pre-trained FR backbones provide superior initialization, we adopt the IR50 weights from their repository as a starting point. However, we address a critical limitation: the massive classification heads designed for thousands of identities lead to rapid overfitting on smaller emotion datasets. We propose replacing these heavy heads with lightweight, attention-based mechanisms to preserve spatial information without parameter bloat.

2.3. Optimization and Generalization Techniques

Training deep networks on heterogeneous, imbalanced datasets requires specialized optimization strategies to ensure convergence and generalization.

- **Warm-up and Learning Rate Schedules:** As demonstrated by Goyal et al. [2], utilizing a learning rate warm-up period is critical for stabilizing training, particularly when using large minibatches. This technique prevents the model from diverging due to large gradient updates during the initial epochs when weights are random.
- **Sharpness-Aware Minimization (SAM):** Standard optimizers like AdamW focus solely on minimizing the training loss value. Foret et al. introduced **SAM** [1], which simultaneously minimizes loss value and loss sharpness. By seeking parameters that lie in "flat minima," SAM significantly improves generalization performance on out-of-distribution data, a crucial property for our multi-source dataset that includes both lab-controlled and in-the-wild images.

3. Methodology

3.1. Data Pre-Processing Methodology

The pre-processing pipeline functions as a sequential execution of image evaluation modules, where images failing specific quality thresholds are programmatically discarded. The evolution of this strategy transitioned from strict curation to selective filtering across two empirical phases.

Phase 1: Aggressive Filtering for a Specialized Use Case The primary hypothesis for Phase 1 was that the model would perform optimally if trained exclusively on perfect, high-quality images that mirrored the target demonstration environment (a user looking directly into a high-quality camera). To enforce this pristine standard, the initial pipeline utilized all available filters sequentially:

- **Lighting Filter:** Analyzed the mean pixel intensity and standard deviation, discarding images that were excessively dark, overly bright, or lacked sufficient contrast.
- **Blur Detection:** Assessed image sharpness using frequency analysis via a Fast Fourier Transform (FFT). Images lacking sufficient high-frequency components were rejected.
- **Roll Correction:** Evaluated the image across four rotational states ($0°, 90°, 180°, 270°$), selecting the orientation that yielded the highest face detection confidence.
- **Yaw Filtering:** RetinaFace detected the face, and 6DRepNet predicted the 3D pose. To guarantee a strict frontal viewpoint, any image with a yaw angle exceeding $45°$ was discarded.
- **Conditional Cropping:** If the ratio of the face area to the total image area was less than 0.4, indicating a small face with significant background, the image was cropped with proportional padding; otherwise, the original image was retained.

In Phase 1, we retained approximately 91,960 images after aggressive filtering.

Phase 2: Resolving the Augmentation Contradiction The aggressively filtered dataset from Phase 1 caused the models to overfit rapidly. To combat this, data augmentation techniques were applied (adding artificial noise, altering brightness, introducing artificial blur, and utilizing MixUp).

This sequence exposed a fundamental contradiction: we were actively removing naturally difficult images during the pre-processing stage, only to artificially recreate those exact conditions during the augmentation stage. Furthermore, augmentations like MixUp produce overlapping visual structures that standard face detectors completely fail to recognize, meaning such images would be universally rejected by strict filters. Recognizing that the model required this data variance to generalize, Phase 2 drastically reduced the aggression of the filtering pipeline:

- **Reduced Aggression:** The universal application of blur, yaw, roll, and lighting filters was halted. For high-quality datasets (AffectNet, CKplus, FERPlus, RAF-DB), the pipeline was stripped down to utilize only structural standardization.
- **Selective Cleaning:** Aggressive techniques were retained only for uncleaned, chaotic datasets (e.g., ExpW and EmoSet), which utilized the cropping module, face detection validation, and manual exclusion of specific emotion labels.

In Phase 2, we started with 310,215 images. After selective pre-processing, we retained 229,478 images.

Standardization and Deduplication To finalize the datasets, all images underwent structural standardization:

- **Channel Normalization:** All inputs were converted to a 3-channel RGB format to ensure consistent tensor depth.
- **Resizing:** Images were zero-padded to achieve a square aspect ratio and subsequently resized to exactly 64×64 pixels. We utilized cubic interpolation if the original image was smaller than 64×64.
- **Deduplication:** Duplicate or highly similar images were purged using a combination of perceptual hashing (pHash) and difference hashing (dHash) to ensure strict separation between training and validation distributions.

Dataset Splitting At the beginning, we followed an 80%/10%/10% split strategy, where 80% of the data was used for training, 10% for validation, and 10% for testing. Towards the end of the project, we adopted a 90%/10% split strategy in most experiments. The training set comprised 90% of the data, while 10% was reserved for validation (evaluation). For the final test set, we exclusively utilized two specific high-quality datasets: CK+ and KDEF. This rigorous separation ensured that our final performance metrics reflected true generalization on standardized benchmarks.

3.2. Architectural Adaptations

3.2.1 ResNet18 with Squeeze-and-Excitation

Our model is based on ResNet18 and is extended with a squeeze-and-excitation (SE) attention module. ResNet18 serves as a stable baseline with moderate capacity (roughly 11–12 million parameters). Results from compact FER literature, including CERN, suggest that attention mechanisms can improve performance across different model sizes, supporting our use of SE as a low-overhead architectural enhancement [**?**]. The input to our network is a standardized RGB face image of size 64×64. Since this resolution is much smaller than the typical ImageNet setup (224×224), we avoid aggressive downsampling in the first layers. In the original ImageNet ResNet, the stem commonly uses a 7×7 convolution with stride 2 followed by max pooling with stride 2. On a 64×64 image, this reduces spatial resolution very early: 64×64 becomes 32×32 after the first stride-2 convolution, and then 16×16 after max pooling, before the main residual stages even begin. This early reduction removes fine-grained facial details (for example small changes around eyes and mouth) that can be important for emotion recognition. The relationship between kernel size, stride, padding, and output resolution follows the standard convolution output sizing rule [**?**]. For this reason, we use a 3×3 convolution with stride 1 in the first layer and omit early pooling, so the model preserves more spatial information in the early feature maps. This is also a common modification when ResNet-style models are adapted to small images (e.g., CIFAR-sized inputs), because preserving spatial detail early tends to improve learning on low-resolution inputs. After the stem, we follow the standard ResNet18 design with four stages of residual blocks and stage-wise downsampling. Residual connections add the block input to the block output, which stabilizes optimization in deep networks [**?**]. After the final stage, global average pooling produces a feature vector that is fed into a dropout-regularized linear classifier to output logits for the six emotion classes. We augment the ResNet18 backbone with squeeze-and-excitation (SE) blocks, which perform lightweight channel-wise attention by learning per-channel scaling factors from globally pooled features, improving representational capacity with minimal overhead [**?**]. The motivation for using a lightweight attention mechanism is consistent with the goal of improving robustness without greatly increasing model size and is aligned with compact FER approaches such as CERN [**?**].

3.2.2 Loss Function and Class Imbalance

Class imbalance is handled through the objective function, meaning the loss function that the model minimizes during training. In standard classification, the model is trained by minimizing cross-entropy loss. With imbalanced data, standard cross-entropy can be dominated by majority classes because they appear more frequently and therefore contribute more gradient updates. To address this, we modify the training objective by using class-balanced focal loss. The class-balanced component assigns larger weights to underrepresented classes using the effective number of samples [**?**], while the focal component reduces the contribution of well-classified (easy) samples so that the model continues learning from harder examples [**?**]. Therefore, we use standard shuffling and address imbalance through the loss rather than through a `WeightedRandomSampler`.

3.3. Explainable AI

AI systems that only expose the input and output of the model to the user are called ”black box models”. The difficulty this design poses lies in the fact that the developer cannot clearly understand the model’s reasoning process. Explainable AI aims to solve this problem by providing a method to visualize the importance of different image regions to the final decision of the model. For out task we experimented with two different methods:

3.3.1 Grad-CAM

The grad-cam method is widely used to explain computer vision models. The algorithm works by analyzing the gradients used by the last convolutional layer and assigning a level of importance for each neuron’s output for the final classification of the image [4]. However this approach did not seem to work as well as we hoped for our model. Applying grad-cam to our model's last convolutional layer resulted in a coarse and inaccurate heatmap. To combat this problem we experimented with using the previous convolutional layers as the base layer for grad-cam which yielded strongly improved results.

The poor performance of the grad-cam method is a result of the heavy downsampling during our model's forward pass. The resolution (8×8) of the final feature maps lacks sufficient spacial information to effectively reconstruct the importance of the low resolution input. This also explains why we achieved better results on a layer using higher resolution feature maps.

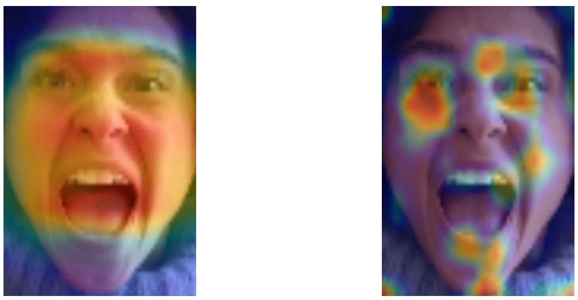

Figure 1. Grad-CAM result of the fourth convolutional layer (left) and the third convolutional layer (right)

3.3.2 Integrated gradients

Integrated gradients is a different algorithm to assess the importance of different image regions for the model's decision. This method works by attributing importance values to the pixels of a baseline image. This image is slowly transformed into the original input image and in each transformation step the gradients of each pixel are calculated. The Importance of each pixel is then set as the average gradient of each pixel, which represents how much the model's output changes based on that piece of information [5]. The result is a fine grained attention map which highlights the most important parts of the image as seen in Figure 2.

This approach seemed interesting to us as the quality of the output did not rely on the architectural details of our model.

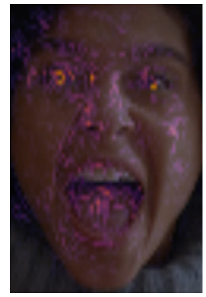

Figure 2. Heatmap resulting from the integrated gradients method

4. Experiments and Results

4.1. Transfer Learning (IR50)

Version 1: Original Full Fine-Tuning The first test used the entire IR50 architecture, including the original classification head.

- **The Problem:** The original head flattened the spatial grid into a 25,088-dimensional vector and sent it into a massive linear layer. This single layer had 12.8 million random parameters, accounting for almost half of the whole model.
- **Result:** Training all 27 million parameters on small datasets caused rapid overfitting. The massive linear layer memorized exact spatial positions, and the high learning rate ruined the pre-trained weights.

Version 2: Layer Freezing and the Memorization Problem

- **Result:** We applied different learning rates, but the network still possessed the oversized 12.8-million-parameter head. This setup led to severe overfitting, shown by a performance gap of over 10 percent between training and validation metrics. The massive parameter count allowed the model to memorize the training data.

Version 3: The Light Head Architecture

- **Result:** With only 5.8 million trainable parameters, a reduced MixUp probability, and an Exponential Moving Average (EMA) decay, this setup successfully stopped the overfitting gap and achieved a peak validation accuracy of 81.38 percent.

4.2. Alternative Architectures Evaluation

During the course of the development process each team member implemented a number of different archtiectures to analyze and compare the performance of different designs. The following section will provide a short overview over the most interesting networks and the results.

4.2.1 Vim (Vision Mamba)

However, Vim is fundamentally designed for high-resolution data, and it proved too difficult to make the architecture focus effectively on the fine details of small facial images.

4.2.2 CCT-7 (Compact Convolutional Transformer)

Even though the architecture was perfectly suited for small images, it simply possessed too few parameters to learn the complex features required for human emotion classification.

4.2.3 ConvNeXt V2 Atto

Similar to CCT-7, the model lacked the parameter count necessary to build robust representations.

4.2.4 GoogLeNet

The GoogLeNet architecture uses a

4.2.5 Other ResNet implementations

Regarding the high accuracy of the ResNet18 + SE implementation ResNet34 and ResNet50 were also trained and evaluated. However the training process was less efficient since the models started overfitting early on due to the high parameter count. As a result the model evaluation yielded a lower accuracy than the original ResNet18 + SE design.

4.3. EfficientNetV2-S Optimization Dynamics

EfficientNetV2-S ultimately emerged as one of the top two best-performing models across all validation benchmarks. To ensure a rigorous comparison, we empirically optimized and tuned each optimizer to find its stable configuration.

Experiment 1: SGD with Nesterov Momentum

- **Result:** Peaked at 74.25 percent validation accuracy.
- **Performance:** SGD [2] proved highly sensitive to hyperparameters. Slight alterations in the learning rate caused massive performance swings, ranging from a complete failure to learn (stagnating below 40 percent accuracy) to achieving a moderate peak of approximately 75 percent.
- **Why it underperformed:** Even at optimal settings, SGD struggled with structural incompatibilities. It applies a uniform learning rate across the network, causing it to under-train depthwise layers while over-training pointwise layers. It lacks the variance tracking required to stabilize the contradictory signals arising from 10 distinct data sources, and it struggles to converge when targets are ambiguous soft labels from Mixup.

Experiment 2: AdamW

- **Result:** Peaked at 83.10 percent validation accuracy.
- **Why it succeeded:** AdamW's adaptive per-parameter learning rates successfully navigated the heterogeneous architecture, balancing updates between depthwise and dense layers. It provided consistent convergence but exhibited mild overfitting in the final epochs.

Experiment 3: SAM + AdamW (Sharpness-Aware Minimization)

- **Result:** Projected to reach 84–86 percent accuracy.
- **Why it succeeded:** SAM [1] improves generalization by minimizing the worst-case loss in the immediate neighborhood of the parameters. This forces the model into flat, stable minima, which is critical for handling the distributional shifts between our 10 diverse datasets. This robustness allowed it to outperform standard AdamW on the most challenging out-of-distribution test sets (e.g., AffectNet).

Optimizer	Peak Validation Accuracy
SGD	74.25%
AdamW	83.10%
SAM + AdamW	**Projected 84–86%**

Table 1. Comparison of optimizer performance on EfficientNetV2-S.

5. Conclusion

This project successfully developed a comprehensive emotion classification system for facial expressions. We implemented a complete pipeline from raw image data through preprocessing to model training and evaluation, utilizing state-of-the-art deep learning techniques.

5.1. Key Contributions

1. **Robust Preprocessing Pipeline**: Developed comprehensive preprocessing procedures including face detection, alignment, deduplication, and quality filtering that significantly improved dataset quality and model robustness.
2. **Attention-Enhanced Architecture**: Integrated Squeeze-and-Excitation blocks into ResNet-18, demonstrating consistent performance improvements over baseline models.
3. **Class-Balanced Training**: Employed focal loss to address class imbalance, resulting in more balanced performance across emotion categories.
4. **Multi-Dataset Evaluation**: Comprehensive evaluation across multiple emotion datasets demonstrated model generalization capabilities and robustness across domain shifts.
5. **Model Interpretability**: Applied Grad-CAM visualization to validate that the model learns emotion-specific facial features, providing insight into the decision-making process.

5.2. Limitations and Future Work

While our system achieves competitive performance, several limitations and opportunities for future work exist:

5.2.1 Limitations

- Performance degradation on low-quality images and extreme face angles
- Difficulty distinguishing between similar emotions (fear/surprise, anger/disgust)
- Limited robustness to occlusions and non-frontal faces

- Dependency on extensive preprocessing and high-quality face detection

5.2.2 Future Directions

- Exploration of transformer-based architectures (Vision Transformers, Compact Transformers) for potentially improved performance
- Integration of temporal information for video-based emotion recognition
- Development of robust multi-task learning approaches combining emotion, age, gender, and identity recognition
- Investigation of domain adaptation techniques to improve cross-dataset generalization
- Incorporation of action unit detection for finer-grained emotion understanding
- Real-time deployment optimization for mobile and edge devices

5.3. Final Remarks

The facial emotion classification system presented in this report demonstrates the effectiveness of combining careful data preprocessing, attention mechanisms, and appropriate loss functions for emotion recognition tasks. The model's interpretability through Grad-CAM analysis confirms that it learns meaningful emotional representations. Future work should focus on improving robustness to challenging conditions and exploring more advanced architecture designs, particularly with emerging transformer-based models. The comprehensive multi-dataset evaluation framework established in this project provides a solid foundation for continued research in this important area of computer vision.

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